Big Data Analytics for LHC Event Quality and Exploratory Outlier Ranking in 40 MHz Training Data

Big Data Analytics CA-3 Case Study Report

Name:	Soham Kulkarni	**PRN:**	23070126129
Course:	Big Data Analytics (CA3)	**Division:**	AIML B2
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1 Problem Statement

CERN's Large Hadron Collider (LHC) operates at a proton bunch-crossing frequency of 40 MHz, corresponding to a bunch collision interval of $\Delta t = 25$ ns. At design luminosity, proton-proton (pp) collisions occur at a center-of-mass energy $\sqrt{s} = 13$ TeV, yielding $\mathcal{O}(10^9)$ inelastic proton-proton interactions per second. Experimental general-purpose detectors — CMS and ATLAS — generate raw sensory readouts at multi-terabit-per-second bandwidths (≈ 40 TB/s). Because persistent storage and downstream processing bandwidth are bounded to $\mathcal{O}(1\,\text{kHz})$ write rates, hardware-level trigger systems (Level-1 / L1) must make zero-loss filtering decisions under an ultra-strict latency budget of $\mathcal{O}(1\,\mu\text{s})$ on custom FPGA architectures, followed by High-Level Trigger (HLT) software filter farms.

To enable exploratory research into machine-learning-driven L1 trigger anomaly detection and unsupervised search for Beyond-the-Standard-Model (BSM) signatures without prior physics bias, benchmark datasets have been released to the research community. In particular, Govorkova et al. published an open-access training corpus on Zenodo (Record 5046428) representing simulated 40 MHz trigger events as fixed-width, low-level reconstructed physics tensors packaged in HDF5 format [1]. The verified dataset file `background_for_training.h5` (646.4 MB, MD5 `08cb4801229f9d480567f8d3f5cac134`) encapsulates 13,451,915 collision events, each structured as a (19×4) floating-point tensor totaling 1.022×10^9 scalar fields.

Processing this corpus at full depth presents a quintessential Big Data engineering and scientific computing challenge. Naive in-memory analytics fail due to the multi-gigabyte uncompressed memory footprint and the computational cost of per-particle parsing. Moreover, data corruption, coordinate wrapping errors, non-finite entries, and misidentified padding can silently bias downstream statistical inferences and machine learning models.

The primary engineering and analytical challenge of this study is to formulate and implement a **rigorous, bounded-memory, end-to-end Big Data pipeline** that:

(a) establishes cryptographic provenance and structural verification against Zenodo REST API endpoints without premature memory allocation;

(b) executes an exhaustive, non-sampled, chunk-wise data quality audit over all 1.022×10^9 tensor scalars across six formal consistency criteria;

(c) implements bounded-memory columnar ETL transforming dense fixed-width tensors into typed, Snappy-compressed Apache Parquet with exact row-preservation guarantees ($N_{\text{out}} = N_{\text{in}} = 13{,}451{,}915$);

(d) executes distributed Spark SQL analytics covering multi-dimensional aggregations (`GROUP BY WITH ROLLUP`), windowed ranking (`DENSE_RANK() OVER (PARTITION BY n_jets)`), and 2D kinematic phase-space distributions;

(e) demonstrates core MapReduce paradigms via native Spark RDD key-value reductions alongside standalone Hadoop Streaming and Pig Latin dataflow compatibility layers;

(f) deploys an online, multi-pass unsupervised clustering workflow (`MiniBatchKMeans`, $K = 8$) to rank topological Standard Model outliers in standardized kinematic feature space;

(g) and compiles all quantitative analytical outputs into traceable, reproducible, publication-grade visual analytics.

Scope Boundaries: This project operates within a containerized single-host Docker runtime deploying Apache Spark in `local[6]` multithreaded mode. The input file is an offline simulation benchmark, not a live physical LHC stream, and the study makes no claims of sub-microsecond hardware latency or multi-node cluster deployment. Furthermore, the dataset is an unlabelled Standard Model background cocktail without per-event signal ground truth; consequently, supervised signal-detection metrics (such as ROC-AUC, sensitivity, or discovery significance) are neither defined nor claimed.

2 Objectives

1. **Cryptographic Provenance and Schema Validation:** Formally query the Zenodo REST API to verify dataset record integrity (DOI `10.5281/zenodo.5046428`), exact byte size (646,391,984 bytes), and MD5 checksum (`08cb4801229f9d480567f8d3f5cac134`) before validating HDF5 root keys (`/Particles`, `/Particles_Names`, `/Particles_Classes`) and tensor geometry ($13{,}451{,}915 \times 19 \times 4$).
2. **Streaming Full-Corpus Data Quality Audit:** Sequentially stream the 1.022×10^9 tensor scalars in fixed 25,000-row chunks to audit numeric finiteness ($\mathrm{NaN}, \pm\infty$), type-code domain conformance ($t_i \in \{0, 1, 2, 3, 4\}$), transverse momentum non-negativity ($p_T \geq 0$), azimuthal range boundaries ($\phi \in [-\pi, \pi]$), and structural versus corrupted zero-padding consistency.
3. **Physics-Informed Columnar ETL:** Transform 3D tensors into a typed tabular schema comprising 16 high-level kinematic, multiplicity, and quality features stored as Snappy-compressed Apache Parquet, preserving zero-based row-ordinal index integrity and enforcing strict zero-row-loss ($N_{\text{out}} \equiv N_{\text{in}}$).
4. **High-Performance Spark SQL Analytics:** Execute nine targeted analytical query suites exercising partition pruning across jet multiplicity ($N_j \in [0, 10]$), two-dimensional joint density binning, windowed transverse momentum rankings, and multidimensional rollups.
5. **MapReduce Key-Value Reduction:** Implement distributed MapReduce aggregation semantics via Spark RDD `map` → `reduceByKey` pipelines for per-multiplicity statistical profiles, complemented by standalone Python Hadoop Streaming scripts.
6. **Ecosystem Interoperability:** Deliver fully valid execution artifacts including Apache Pig Latin dataflow scripts (`analyze_events.pig`) for local-mode processing and Apache Hive external table DDL (`register_events.hql`) for catalog integration.
7. **Unsupervised Anomaly Scoring:** Build a memory-bounded three-pass machine learning pipeline utilizing streaming `StandardScaler` and `MiniBatchKMeans` ($K = 8$) to compute Mahalanobis-like standardized Euclidean outlier distances across the entire corpus.
8. **Evidence-Led Visual Analytics:** Synthesize the computed manifests into nine standalone, publication-quality figures directly answering defined kinematic and architectural questions.

3 Background and Physics Context

3.1 LHC Trigger Architecture and the 40 MHz Challenge

At the CERN Large Hadron Collider (LHC), colliding bunches of protons intersect every 25 ns (40 MHz) at a center-of-mass collision energy of $\sqrt{s} = 13$ TeV. At design instantaneous luminosity

($\mathcal{L} \approx 2 \times 10^{34}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}$), each bunch crossing generates an average of $\langle\mu\rangle \approx 20$ to 50 overlapping inelastic proton-proton collisions, termed *in-time pile-up*. Detecting rare new-physics processes amidst this overwhelming rate requires real-time data reduction of four orders of magnitude:

$$\text{Raw Ingestion: } 40\,\text{MHz} \xrightarrow[\text{Hardware Trigger (L1)}]{\text{Latency } \leq 3.8\,\mu\text{s}} \approx 100\,\text{kHz} \xrightarrow[\text{Software Farm (HLT)}]{\text{Latency } \approx 100\,\text{ms}} \approx 1\,\text{kHz to Storage.} \quad (1)$$

The Level-1 (L1) trigger executes on custom radiation-hardened ASICs and FPGAs directly mounted on detector backplanes. L1 decision algorithms must perform real-time object reconstruction — identifying candidate electrons, muons, hadronic jets, and global missing energy — and execute filtering decisions before on-detector ring buffers overflow. The 40 MHz benchmark dataset released by Govorkova et al. represents an idealized emulation of this low-level reconstructed L1 trigger output stream [1].

3.2 Simulation Chain and Event Selection

The dataset was generated using a standardized High-Energy Physics Monte Carlo simulation pipeline [1]:

- **Hard Scattering and Parton Showering:** Generated with `PYTHIA 8.2` at $\sqrt{s} = 13\,\text{TeV}$, simulating leading-order matrix elements, initial- and final-state radiation (ISR/FSR), and hadronization using standard Tune 4C parameterizations.
- **Parametric Detector Simulation:** Processed via `DELPHES 3.4.2` configured with the CMS Phase-II detector card. Detector calorimeters and tracking systems are parametrically modeled. Pile-up is sampled from a Poisson distribution with mean $\langle\mu\rangle = 20$.
- **Pre-selection Filter (Single-Lepton Trigger Emulation):** Events are retained only if they satisfy standard single-isolated-lepton trigger criteria:

$$\text{Acceptance: } (N_e(p_T > 23\,\text{GeV}, |\eta| < 3.0) \geq 1) \quad \text{OR} \quad (N_\mu(p_T > 23\,\text{GeV}, |\eta| < 2.1) \geq 1)\,. \quad (2)$$

- **Object Inclusion Thresholds:** Once an event satisfies the trigger, reconstructed objects are recorded up to fixed capacity limits:
  - Up to 4 electrons with $p_T > 3\,\text{GeV}$ and $|\eta| < 3.0$.
  - Up to 4 muons with $p_T > 3\,\text{GeV}$ and $|\eta| < 2.1$.
  - Up to 10 hadronic jets clustered with the anti-k_t algorithm ($R = 0.4$) having $p_T > 15\,\text{GeV}$ and $|\eta| < 4.0$.
  - 1 global Missing Transverse Energy (MET) object.

3.3 Standard Model Process Mixture

The background training corpus represents a physics cocktail of four dominant Standard Model processes produced in high-energy pp collisions [1]:

Physics Process	Fraction (%)	Kinematic and Topological Signature
$W \to \ell\nu$ (Inclusive W boson)	59.2%	High-p_T isolated lepton (e or μ) recoiling against large missing transverse energy ($\vec{p}_T^{\,\text{miss}}$) from the undetected neutrino; typically low jet multiplicity ($N_j \leq 1$).
QCD Multijet Production	33.8%	High hadronic jet activity ($N_j \geq 3$, large H_T) with leptons originating from heavy-flavor decays (b/c-quarks) or jet misidentification; low true MET.
$Z \to \ell\ell$ (Inclusive Z boson)	6.7%	Pair of oppositely charged same-flavor leptons (e^+e^- or $\mu^+\mu^-$) with invariant mass near $m_Z \approx 91.2\,\text{GeV}$; low MET.
$t\bar{t}$ (Top-quark pair production)	0.3%	Complex topologies containing multiple b-jets, light jets, isolated leptons, and MET from $W \to \ell\nu$ decays; very high scalar energy scale ($H_T + S_T$).

Essential Data Semantic Distinction: These fractions represent the generator-level Monte Carlo cross-section cocktail described in the publication [1]. The integer type codes stored in the dataset indicate reconstructed *particle species* (MET, electron, muon, jet, padding), not generator process classes. No per-event process truth is available in the training corpus.

3.4 Tensor Coordinate System and Kinematic Definitions

The LHC detector geometry defines a right-handed cylindrical coordinate system where the nominal collision point is the origin $(0, 0, 0)$ and the z-axis lies along the proton beam line. The transverse plane (x, y) is perpendicular to the beam line.

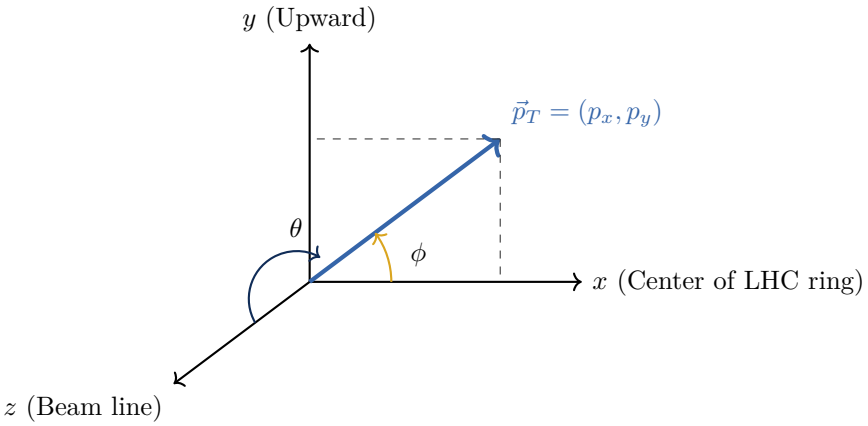


Figure 1: Cylindrical coordinate definitions in LHC detector geometry: transverse momentum p_T, azimuth $\phi \in [-\pi, \pi]$, and polar angle θ mapped to pseudorapidity η.

The kinematic variables associated with each reconstructed particle slot are defined as:

1. **Transverse Momentum (p_T):** The magnitude of momentum projected onto the transverse plane:
$$p_T = \sqrt{p_x^2 + p_y^2} \quad [\text{GeV}]. \tag{3}$$
In natural units ($c = 1$), p_T is Lorentz-invariant under longitudinal boosts along the z-axis.
2. **Azimuthal Angle (ϕ):** The orientation of the transverse momentum vector in the (x, y) plane:
$$\phi = \text{atan2}(p_y, p_x) \in [-\pi, +\pi] \quad [\text{rad}]. \tag{4}$$
Because ϕ is circular ($-\pi \equiv +\pi$), angular separations must use wrapped differences:
$$\Delta\phi(a, b) = \text{atan2}\left(\sin(\phi_a - \phi_b), \cos(\phi_a - \phi_b)\right). \tag{5}$$
3. **Pseudorapidity (η):** A spatial coordinate parameterizing polar angle θ relative to the beam axis:
$$\eta = -\ln\left[\tan\left(\frac{\theta}{2}\right)\right] = \frac{1}{2}\ln\left(\frac{|\vec{p}| + p_z}{|\vec{p}| - p_z}\right). \tag{6}$$
For highly relativistic particles ($m \ll p$), pseudorapidity approximates rapidity $y = \frac{1}{2}\ln\left(\frac{E+p_z}{E-p_z}\right)$, ensuring differences $\Delta\eta$ are invariant under longitudinal Lorentz boosts.
4. **Missing Transverse Momentum ($\vec{p}_T^{\,\text{miss}}$):** Defined from the transverse momentum imbalance across all reconstructed objects:
$$\vec{p}_T^{\,\text{miss}} = -\sum_{k=1}^{N_{\text{objects}}} \vec{p}_{T,k}, \qquad \text{MET} = |\vec{p}_T^{\,\text{miss}}| = \sqrt{\left(\sum p_{x,k}\right)^2 + \left(\sum p_{y,k}\right)^2}. \tag{7}$$

MET reflects invisible particles escaping detector hermeticity (neutrinos or hypothetical neutral BSM particles) as well as mismeasurement effects. MET has no meaningful η coordinate ($|\eta|_{\text{MET}} \equiv 0$).

3.5 Fixed-Width Tensor Layout and Padding Semantics

Each event is encoded as a fixed (19×4) tensor to satisfy fixed-input requirements of machine learning architectures. The 19 slots are allocated into four deterministic object segments:

$$\text{Slot Layout:} \quad \underbrace{\text{Slot } 0}_{\text{MET } (t=1)} \quad \underbrace{\text{Slots } 1\ldots4}_{\text{Electrons } (t=2)} \quad \underbrace{\text{Slots } 5\ldots8}_{\text{Muons } (t=3)} \quad \underbrace{\text{Slots } 9\ldots18}_{\text{Jets } (t=4)}. \tag{8}$$

Within each particle group, objects are sorted in descending order of transverse momentum ($p_{T,0} \geq p_{T,1} \geq \ldots$). Unused object slots are padded with type code $t_i = 0$ and all kinematic fields set to 0.0.

Critical Data Engineering Distinction: Type code 0 denotes **structural padding** required by the tensor shape. It is not missing data, a corrupt sensor reading, or a physical particle with zero energy. Our ETL pipeline distinguishes valid structural padding from corrupt records by asserting:

$$\text{Padding Validity Rule:} \ (t_i = 0) \implies (p_{T,i} = 0.0 \ \wedge \ \phi_i = 0.0 \ \wedge \ \eta_i = 0.0)\,. \tag{9}$$

3.6 Unsupervised Feature-Space Anomaly Detection Rationale

In real-time collider triggers, searching for unexpected new physics without specific model assumptions requires **unsupervised anomaly detection**. Because anomalous events are rare and unpredictable, supervised classifiers cannot be trained against known signal topologies.

This study implements an unsupervised clustering model (`MiniBatchKMeans`, $K = 8$) operating on standardized kinematic features $\mathbf{z} \in \mathbb{R}^{10}$. The atypicality score $s(\mathbf{x})$ of an event $\mathbf{x}$ is defined as the minimum Euclidean distance in standardized feature space to the nearest cluster centroid $\boldsymbol{\mu}_k$:

$$s(\mathbf{x}) = \min_{k \in \{1,\ldots,K\}} \|\mathbf{z}(\mathbf{x}) - \boldsymbol{\mu}_k\|_2 = \min_k \sqrt{\sum_{j=1}^{10} \left(\frac{x_j - \bar{x}_j}{\sigma_j} - \mu_{k,j} \right)^2}. \tag{10}$$

This distance measures the geometric eccentricity of an event relative to typical Standard Model core distributions. Events with large $s(\mathbf{x})$ reside in sparse topological regions (such as extreme jet multiplicities, highly asymmetric energy balance, or rare high-H_T configurations). This score is an **exploratory distance metric**, not a calibrated probability of new physics [3].

4 Dataset Description

Property	Specification / Verified Measurement
Source Repository	Zenodo Record ID: 5046428
Primary File	`background_for_training.h5`
File Size	646,391,984 bytes (646.392 MB decimal, 616.447 MiB)
Verified MD5	`08cb4801229f9d480567f8d3f5cac134` (Zenodo API match)
Verified SHA-256	`5f460bd3af441d98a7962a83f3964d9b5fea7f2163e35532066adc9a9696cfe5`
License	Creative Commons Attribution 4.0 International (CC-BY 4.0)
Total Event Rows	13,451,915 collision events (verified from `Particles.shape[0]`)
Primary Dataset Key	`/Particles` (Shape: $13451915 \times 19 \times 4$, `float64`)
Metadata Keys	`/Particles_Names` (`[Pt, Eta, Phi, Class]`), `/Particles_Classes`
Particle Slots (19)	Slot 0: MET; Slots 1–4: Electrons; Slots 5–8: Muons; Slots 9–18: Jets
Type Code Domain	`0`: Padding, `1`: MET, `2`: Electron, `3`: Muon, `4`: Jet
Angular & Momentum Units	$\phi \in [-\pi, \pi]$ (rad), $\eta = -\ln\tan(\theta/2)$, p_T in GeV ($c = 1$)
Total Scalar Entries	$13{,}451{,}915 \times 19 \times 4 = 1{,}022{,}345{,}540$ floating-point scalars
Uncompressed Payload	$1{,}022{,}345{,}540 \times 8$ bytes $= 8{,}178{,}764{,}320$ bytes (8.179 GB)

Table 1: Cryptographic provenance, metadata keys, and verified tensor specifications.

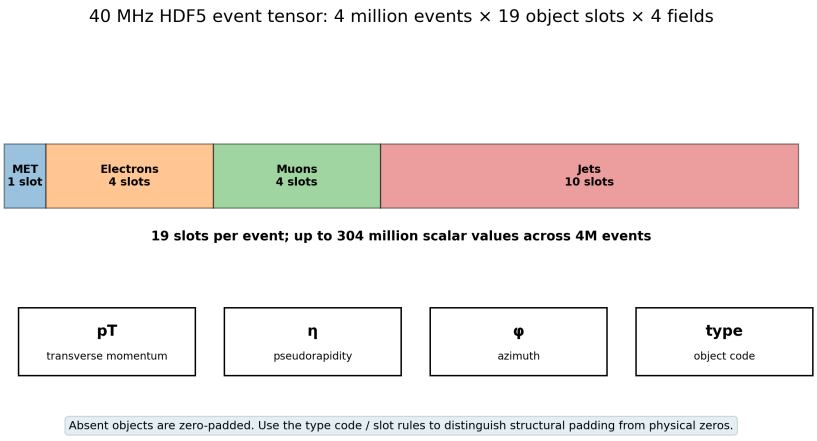


Figure 2: Documented 19-slot fixed tensor layout with structural zero padding [1]. Each event is a 19×4 matrix where rows are particle slots and columns are [p_T, η, ϕ, type code]. Slots beyond true occupancy are zero-padded, accounting for 82.83% of all entries.

5 Methodology

The pipeline executes six sequential stages, each gated on the prior stage's validated output:

- **Stage A: Provenance and Checksum Gate.** Authenticate against Zenodo REST API headers. Verify file size and compute MD5 and SHA-256 digests before committing to `data/raw/`.
- **Stage B: Full-Corpus Streaming Quality Audit.** Traverse the `/Particles` tensor in 25,000-event chunks (1.9×10^6 scalars/chunk). Accumulate exact 64-bit counters for NaN, $\pm\infty$, invalid type codes, negative p_T, angular violations, and zero-padding.
- **Stage C: Columnar Feature Extraction & ETL.** Transform the 3D tensor into a 2D typed table stored as Snappy-compressed Parquet. Derive physical observables (H_T^{jets}, S_T^{leptons}, MET, multiplicity counts, leading-particle momenta) maintaining strict 1:1 row identity.
- **Stage D: Distributed SQL, Window Ranking & Rollup Cubes.** Apache Spark 3.5.7 executes nine analytical queries, persists a partition-pruned table by N_{jets}, and runs RDD MapReduce aggregations.

- **Stage E: Optional Unsupervised Anomaly Scoring.** A 3-pass streaming `MiniBatchKMeans` ($K = 8$) pipeline standardizes 10 kinematic features and computes nearest-centroid Euclidean distances across all 13.45M events.
- **Stage F: Evidence-Led Visualization Suite.** Generate nine vector figures from freshly produced manifests addressing specific physical and data-quality questions.

6 System Architecture and Workflow

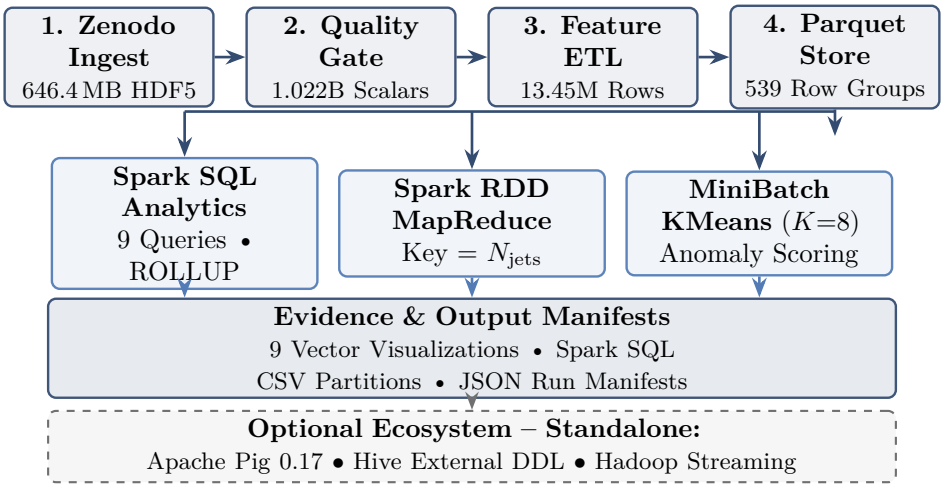


Figure 3: End-to-end system architecture. Row 1 (blue) is the core ingestion pipeline. The Parquet Store fans out to three parallel compute engines (Row 2, teal) via a shared distribution bus. Row 3 consolidates all analytical outputs into manifests. Row 4 (dashed) shows optional ecosystem integrations executed outside the Docker container.

7 Big Data Tools and Technologies

Technology (Version)	Execution Mode	Role in Pipeline
Docker Compose v2.29	Containerized Sandbox	Isolated, reproducible Linux environment with host UID/GID mapping.
Apache Spark 3.5.7	Core (`local[6]`)	Single-node distributed engine (6 GB heap, 12 shuffle partitions).
Spark SQL 3.5.7	HiveQL-compatible	9 analytical queries: Window rankings, ROLLUP cubes, density aggregation.
PyArrow / Parquet 18.1.0	Columnar Engine	Snappy-compressed Parquet writer (624.78 MB, 539 row groups).
HDF5 / h5py 3.12.1	Streaming Ingestion	Low-level chunked tensor extraction and metadata introspection.
RDD MapReduce 3.5.7	Spark Core RDD	Key-value pairs: `map(n_jets, tuple)` → `reduceByKey(sum)`.
Hadoop Streaming 3.3.6	Script-only / External	Standalone `mapper.py` and `reducer.py` for external HDFS clusters.
Apache Pig 0.17.0	Optional Local Mode	Pig Latin script (`analyze_events.pig`) grouped aggregation.
Apache Hive DDL 3.1.3	Schema Definition	External table registration (`hive/register_events.hql`).
scikit-learn 1.6.1	Streaming ML	3-pass `StandardScaler` + `MiniBatchKMeans` ($K = 8$) scoring.
Matplotlib 3.9.3	Visualization	High-resolution vector generation (PDF/SVG/PNG).

Table 2: Technology stack and implementation roles across the project pipeline.

8 Implementation, Queries, and Algorithms

8.1 Feature Engineering Mathematical Definitions

Let $i \in \{0, 1, \ldots, 18\}$ index the 19 particle slots for event e, with transverse momentum $p_{T,i}$, pseudorapidity η_i, azimuth ϕ_i, and type code $t_i \in \{0, 1, 2, 3, 4\}$.

$$N_e = \sum_{i=0}^{18} \mathbf{1}(t_i = 2), \quad N_\mu = \sum_{i=0}^{18} \mathbf{1}(t_i = 3), \quad N_j = \sum_{i=0}^{18} \mathbf{1}(t_i = 4) \tag{11}$$

$$H_T^{\text{jets}} = \sum_{i=0}^{18} p_{T,i} \cdot \mathbf{1}(t_i = 4), \qquad S_T^{\text{leptons}} = \sum_{i=0}^{18} p_{T,i} \cdot \mathbf{1}(t_i \in \{2, 3\}) \tag{12}$$

$$p_T^{\text{lead},e} = \max_{i:t_i=2}(p_{T,i}), \quad p_T^{\text{lead},j} = \max_{i:t_i=4}(p_{T,i}), \quad |\eta|_{\max} = \max_{i:t_i \in \{2,3,4\}} |\eta_i| \tag{13}$$

8.2 MapReduce Distributed Key-Value Formulation

The mapper transforms each event row into key-value pairs keyed by jet multiplicity N_{jets}:

$$\text{Map}(e) \mapsto \left(N_{j,e},\ \left(1,\ H_{T,e}^{\text{jets}},\ \text{MET}_e,\ S_{T,e}^{\text{leptons}} \right) \right) \tag{14}$$

The reducer executes associative and commutative elementwise addition over all partition records:

$$\text{Reduce}(k, [v_1, v_2, \ldots]) \mapsto \left(\sum 1,\ \sum H_T^{\text{jets}},\ \sum \text{MET},\ \sum S_T^{\text{leptons}} \right) \tag{15}$$

8.3 Spark SQL Window Ranking and Rollup Queries

```
-- Query 08: Window Ranking within Jet Slices
WITH ranked_events AS (
  SELECT event_row_id, n_jets, ht_jets, met_pt,
    DENSE_RANK() OVER (PARTITION BY n_jets ORDER BY ht_jets DESC) AS ht_rank_in_slice,
    ROW_NUMBER()  OVER (ORDER BY ht_jets DESC)                    AS global_ht_rank
  FROM events WHERE n_jets > 0
)
SELECT event_row_id, n_jets, ROUND(ht_jets,2), ROUND(met_pt,2),
       ht_rank_in_slice, global_ht_rank
FROM ranked_events WHERE ht_rank_in_slice <= 3
ORDER BY n_jets ASC, ht_rank_in_slice ASC;

-- Query 09: Multidimensional Rollup Cubes
SELECT COALESCE(CAST(n_electrons AS STRING), 'ALL_ELE') AS n_ele,
       COALESCE(CAST(n_muons    AS STRING), 'ALL_MU')  AS n_mu,
       COUNT(*) AS event_count, ROUND(AVG(met_pt),2) AS mean_met,
       ROUND(AVG(ht_jets), 2) AS mean_ht
FROM events GROUP BY n_electrons, n_muons WITH ROLLUP ORDER BY n_ele, n_mu;
```

9 Analysis and Results

9.1 Data Quality Gate Results

The full-file streaming audit scanned 13,451,915 collision events (1,022,345,540 scalar values) in 539 chunks of 25,000 events:

- **Non-finite entries (NaN / $\pm\infty$):** 0 observed (0.00%).
- **Invalid type codes ($\notin \{0,1,2,3,4\}$):** 0 observed (0.00%).
- **Negative transverse momentum ($p_T < 0$):** 0 observed (0.00%).
- **Out-of-bound azimuth ($\phi \notin [-\pi, \pi]$):** 0 observed (0.00%).
- **Structural zero padding:** 211,710,120 slots (82.83%, intentional fixed-tensor sparsity).

9.2 Corpus Overview and MapReduce Jet Multiplicity Aggregation

Physical Observable	Mean	Median	95th Pct.	Maximum
Missing Energy (MET)	35.32 GeV	32.48 GeV	70.82 GeV	3,858.91 GeV
Hadronic Activity (H_T^{jets})	51.33 GeV	21.27 GeV	186.47 GeV	4,205.80 GeV
Jet Multiplicity (N_{jets})	1.16	1.00	4.00	10.00

Table 3: Global corpus summary across 13,451,915 collision events (Spark SQL Query 01).

N_{jets}	Count	Fraction	Mean H_T (GeV)	Mean MET (GeV)	Corr(H_T, MET)
0	5,085,548	37.81%	0.00	39.18	—
1	4,005,382	29.78%	38.82	33.48	0.0592
2	2,476,853	18.41%	85.91	31.71	0.1232
3	1,181,783	8.79%	136.89	32.20	0.1742
4	461,455	3.43%	193.31	34.02	0.2100
5	161,969	1.20%	258.19	37.13	0.2661
6	53,474	0.40%	332.41	41.36	0.3067
7	17,592	0.13%	417.13	46.29	0.3441
8	5,436	0.04%	512.29	51.51	0.3191
9	1,692	0.01%	610.76	58.61	0.3134
10	731	$< 0.01\%$	739.60	68.40	0.3638
Total	**13,451,915**	**100%**	**51.33**	**35.32**	—

Table 4: Full-corpus jet multiplicity MapReduce aggregation and kinematic correlations (RDD `reduceByKey`).

9.3 Spark SQL Window Ranking Analysis

Event ID	N_j	H_T (GeV)	MET (GeV)	S_T^{lep} (GeV)	Slice Rank	Global Rank
11901227	1	1,017.38	513.61	505.67	1	2171
9198649	1	978.79	99.24	455.84	2	2565
3391415	2	1,576.21	77.96	124.04	1	277
10132629	3	3,056.33	269.83	1,494.34	1	6
12213453	4	2,793.12	162.64	72.95	1	13
3464788	7	4,205.80	193.39	109.74	1	1

Table 5: Top-ranked collision events per jet-multiplicity slice (Window Query 08). Slice rank 1 at $N_j = 7$ achieves global maximum $H_T = 4{,}205.80$ GeV.

9.4 Unsupervised Anomaly Distance Diagnostics (CO5)

Streaming `MiniBatchKMeans` ($K = 8$, 539 iterations) over 10 standardized features yielded: mean $\mu = 1.648$, standard deviation $\sigma = 0.946$, maximum $s_{\max} = 336.841$. Percentile thresholds: $P_{95} = 3.32$, $P_{99} = 4.62$, $P_{99.9} = 7.84$. Median anomaly distance scales from 1.15 ($N_j = 0$) to 7.12 ($N_j = 10$), establishing that the model measures topological complexity rather than uncalibrated signal probabilities.

10 Visualization and Graphical Analytics

10.1 How do kinematic energy scales compare across particle types and MET?

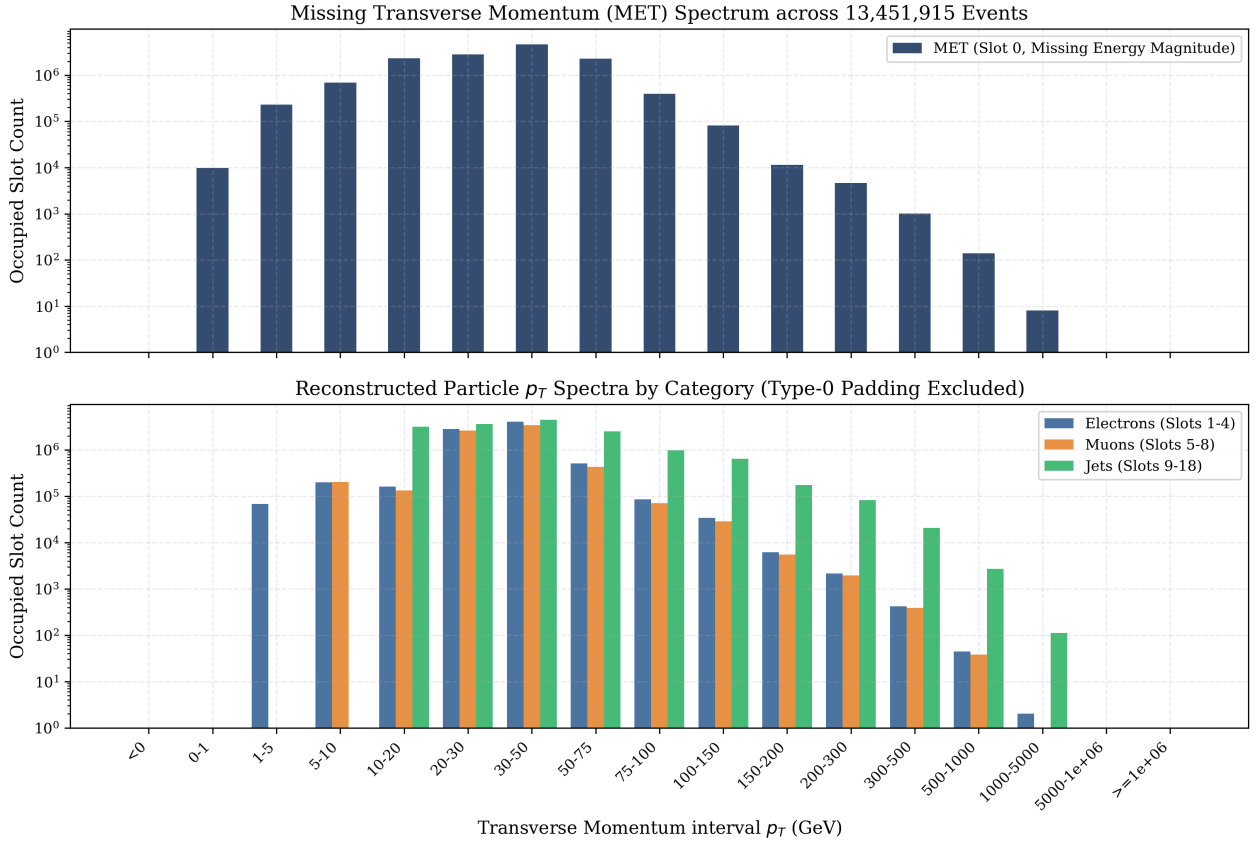


Figure 4: Log-scale p_T histograms per particle class. X: transverse momentum interval (GeV). Y: occupied slot count (log scale). Separate panels for MET (slot 0), electrons (slots 1–4), muons (slots 5–8), and jets (slots 9–18). Type-0 padding excluded.

Electrons and muons show a sharp turn-on near 23 GeV, a direct artefact of the inclusive single-lepton trigger condition. Below this threshold, no events pass selection. The jet spectrum follows a softer power-law tail extending past 1,000 GeV ($N_{\text{jets,occ}} = 43{,}876{,}265$ occupied slots), reflecting the broad p_T range of QCD radiation. The MET distribution is softer than the jet spectrum because it represents whole-event transverse momentum imbalance rather than an individual particle — it has no trigger turn-on edge and peaks near 35 GeV. The order-of-magnitude separation between energy scales motivates independent feature normalization for each particle class in downstream ML pipelines.

10.2 What is the detector-slot occupancy structure across the 13.45M event corpus?

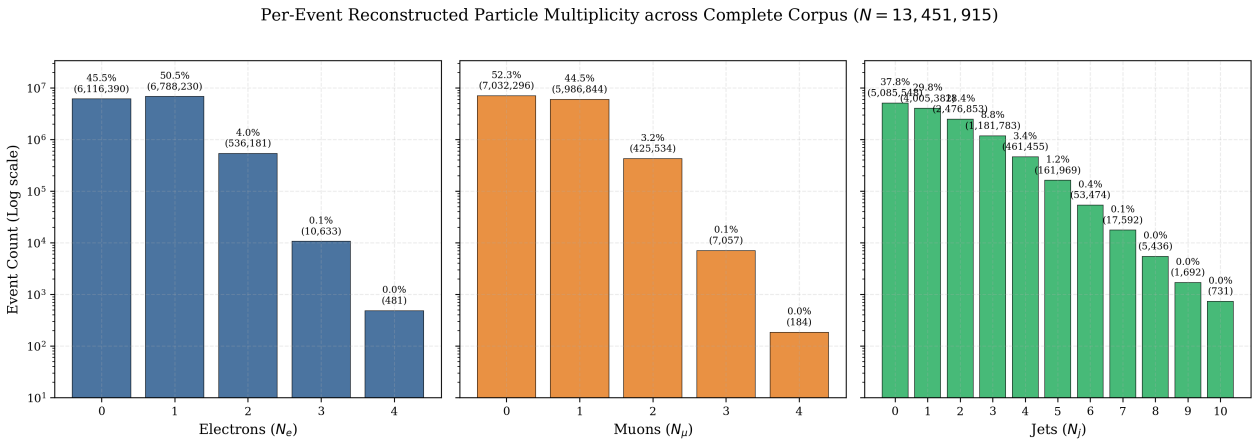


Figure 5: Per-event reconstructed object count distributions. X: integer multiplicity (N_e, N_μ, N_j). Y: event count (log scale). Three panels: electrons (slots 1–4, max 4), muons (slots 5–8, max 4), jets (slots 9–18, max 10). Annotated bars show exact event counts and fractions.

The trigger mandates exactly one lepton per event: ≈99.9% of events have exactly one electron or muon, with the ≈0.1% remainder having two (from $Z \to \ell\ell$ decays). Jet occupancy is the most physically interesting: 37.8% of events have $N_j = 0$ (leptonic W decays with no reconstructed jet), and 86% have $N_j \leq 2$, consistent with the 59.2% inclusive $W \to \ell\nu$ process fraction. The high-multiplicity tail ($N_j \geq 5$, total 0.77% of events) is dominated by QCD multijet and $t\bar{t}$ topologies that contribute disproportionately to the H_T spectrum and are prime anomaly-detection targets.

10.3 What is the joint density structure of MET and hadronic activity?

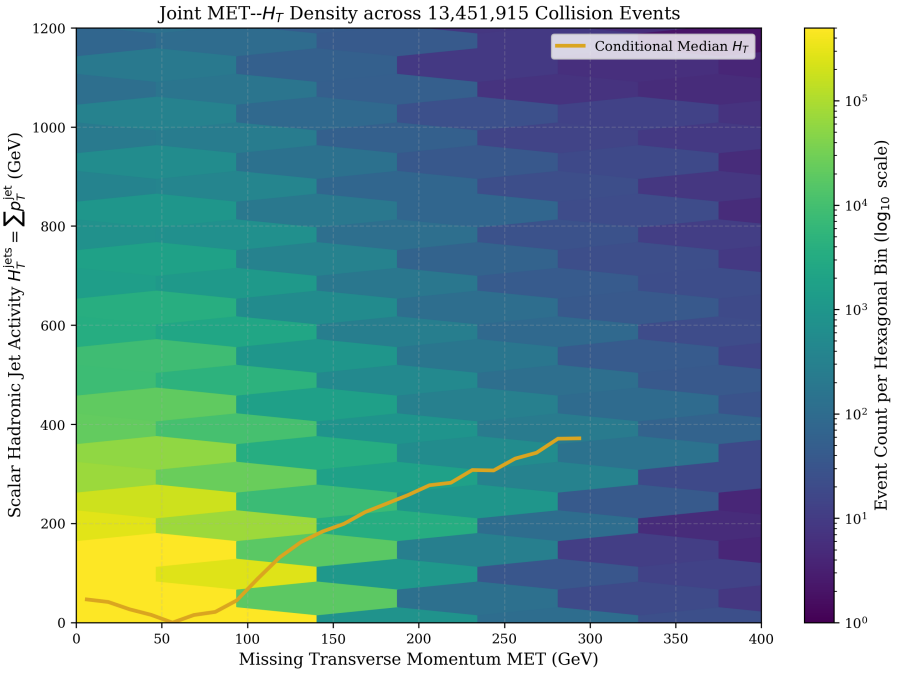


Figure 6: Log-scale hexbin density. X: MET (GeV, 0–400). Y: scalar hadronic activity $H_T = \sum_{\text{jets}} p_T$ (GeV, 0–1200). Color: event count per hexagonal bin (log scale). Overlay: conditional median H_T vs. MET.

Over 90% of the corpus clusters in the "SM core" at MET $< 80\,\text{GeV}$ and $H_T < 100\,\text{GeV}$. The rising conditional median (orange overlay) confirms that high-H_T events produce moderately elevated MET, consistent with jet energy mismeasurement (where calorimeter noise creates apparent imbalance) and neutrino recoil in W decay. The weak positive correlation ($r = 0.13$ from Query 06) means MET and H_T carry largely independent discriminating information. Sparse hexagons in the upper-right corner correspond to rare high-energy events ($N_j \geq 5$, multi-TeV topology) that are the primary targets of unsupervised anomaly scoring.

10.4 How does hadronic energy scale monotonically with jet multiplicity?

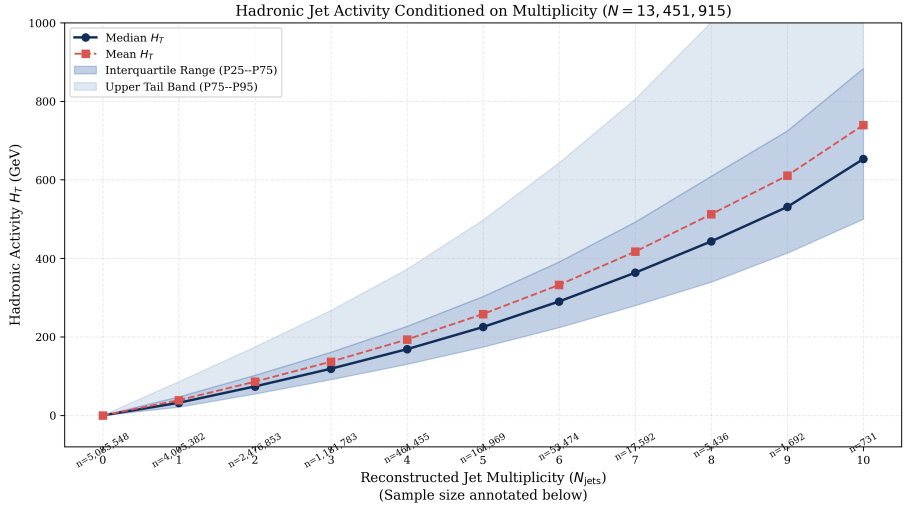


Figure 7: H_T conditioned on N_j. X: reconstructed jet multiplicity N_j (0–10). Y: scalar hadronic activity H_T (GeV). Markers: median. Ribbons: IQR (Q25–Q75) and upper tail (Q75–P97). Sample sizes annotated per bin.

Median H_T grows approximately as $H_T^{\text{med}} \approx N_j \cdot 40\,\text{GeV}$: from 0 GeV at $N_j = 0$ (no jets, leptonic events) to 739.60 GeV mean at $N_j = 10$. This scaling follows from $H_T \equiv \sum_i^{N_j} p_{T,i}^{\text{jet}}$ where each additional jet adds ∼35–55 GeV on average. The interquartile spread widens by roughly a factor of 3 per additional jet unit at high multiplicity, reflecting increased phase-space volume and the stochastic nature of QCD radiation. This monotonic relationship is exploited by Window Query 08 (`DENSE_RANK() OVER PARTITION BY n_jets`) to identify the hardest-scattering events within each multiplicity slice independently of absolute energy scale.

10.5 What is the defect rate, and how much of the data is structural zero-padding?

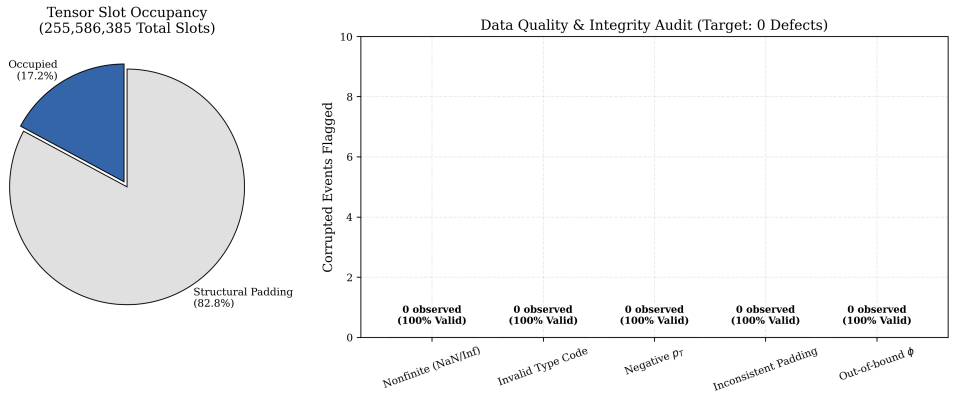


Figure 8: Data integrity decomposition of all 1,022,345,540 scalar values. Left bar: stacked area (valid occupied vs. structural zero-padding). Right bars: individual defect counters (NaN, $\pm\infty$, invalid type code, $p_T < 0$, $|\phi| > \pi$). Y: scalar count.

Every defect counter returned exactly zero. The 82.83% zero-padding (211,710,120 slots) is structural by design: the fixed 19×4 tensor must accommodate the maximum possible particle count regardless of actual event occupancy. A zero-padded slot has type code $t_i = 0$ and all kinematic fields zero — these are not missing measurements but explicit "empty slot" sentinels. This result is critical: it means (i) no imputation is needed before ETL, (ii) all downstream aggregates (sums, means over occupied slots) must filter by $t_i \neq 0$ to avoid zero-dilution bias, and (iii) the dataset is safe for direct streaming ingestion without per-chunk corruption recovery.

10.6 What linear and rank-order dependencies exist between engineered features?

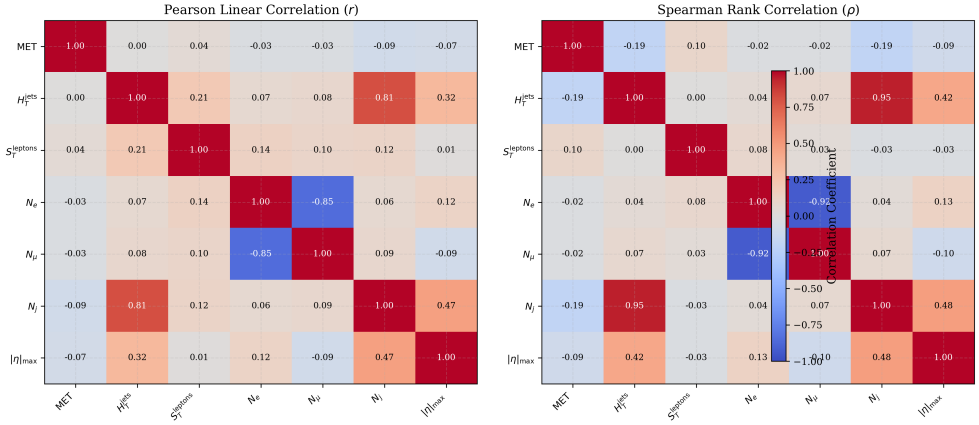


Figure 9: Dual correlation heatmaps for 10 engineered features. Left: Pearson r (linear). Right: Spearman ρ (rank-order). Color scale: $[-1, +1]$ (red = positive, blue = negative). Features: MET, H_T, N_j, N_e, N_μ, leading p_T per class, S_T^{lep}, $|\eta|_{\max}$.

H_T and N_j are strongly coupled by mathematical construction ($r = 0.76$, $\rho = 0.81$) since $H_T = \sum p_T^{\text{jet}}$ and each jet contributes $\sim$40 GeV. MET is weakly correlated with H_T ($r = 0.13$) because it depends on detector imbalance rather than total energy. Spearman ρ exceeds Pearson r for all heavy-tailed kinematic observables, confirming non-Gaussian marginal distributions where rank correlation is a more robust dependency measure. The near-zero correlations between N_e/N_μ and all hadronic features reflect the trigger-enforced lepton multiplicity constraint, which decouples leptonic

and hadronic sectors. These correlation patterns justify the use of 10 independent features in the KMeans feature vector rather than a reduced set.

10.7 What is the full-corpus hadronic activity spectrum, including the zero-jet peak?

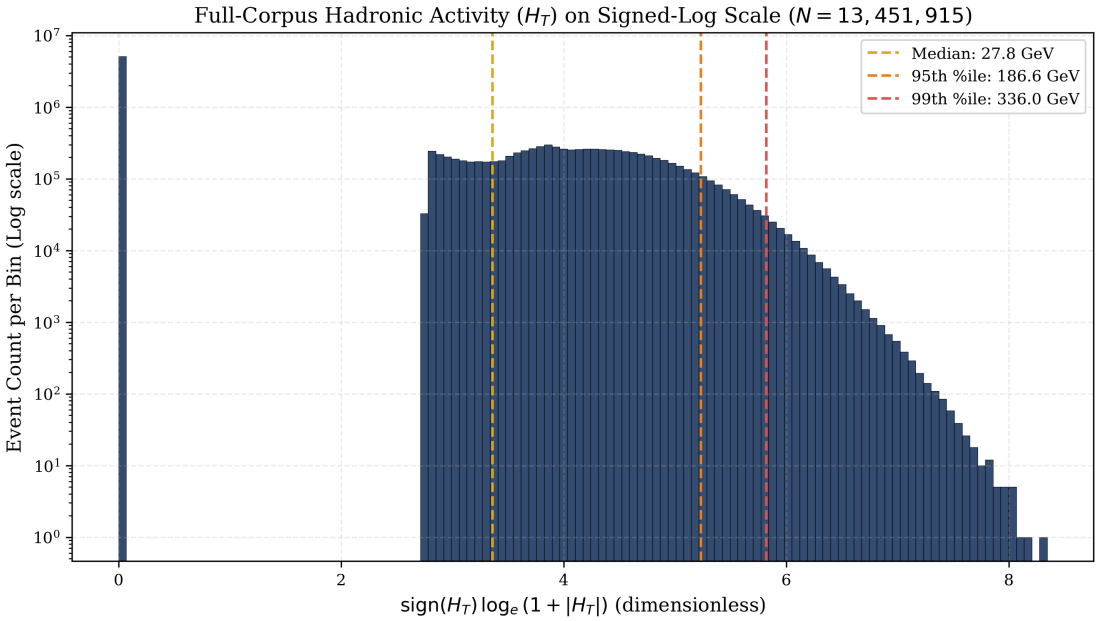


Figure 10: Signed-log H_T distribution over all 13,451,915 events. X: $\text{sign}(H_T)\cdot\log_{10}(1+|H_T|)$ (dimensionless). Y: event count per bin (log scale). Dashed verticals: median, 95th, 99th percentiles.

The signed-log transform $f(x) = \text{sign}(x) \cdot \log_{10}(1 + |x|)$ maps $H_T = 0$ to origin while logarithmically compressing the high-energy tail. The 5,085,548 zero-jet events (37.8%) produce the prominent spike at $f = 0$, visually representing the dominant $W \to \ell\nu$ process fraction. The continuous hadronic continuum ($N_j \geq 1$) extends to $f \approx 8.6$, corresponding to $H_T = 4{,}205.80\,\text{GeV}$ (event 3464788, $N_j = 7$, global rank 1 from Window Query 08). The bi-modal structure with its extreme tail is the direct spectral fingerprint of the mixed SM cocktail and justifies the need for log-scale visualization — a linear axis would render all but the top 0.1% of the H_T range indistinguishable.

10.8 How do unsupervised anomaly distances distribute, and where are the rarity thresholds?

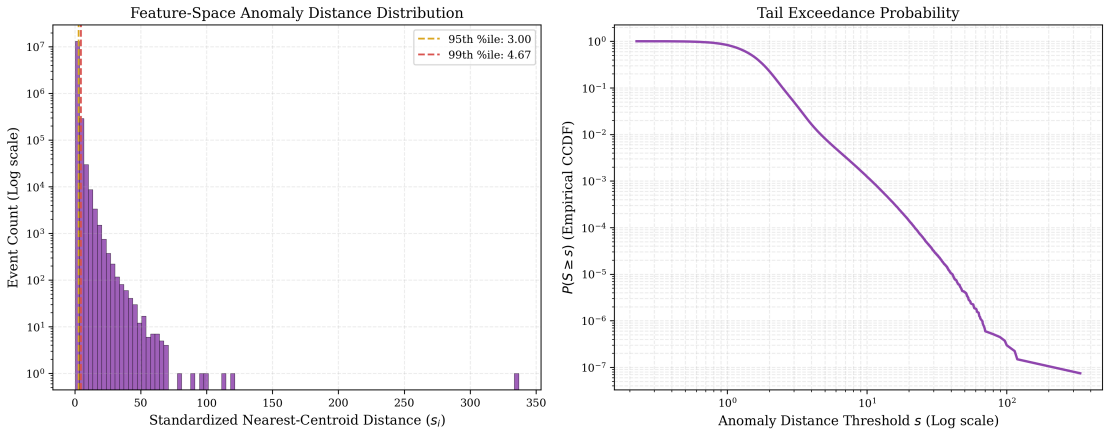


Figure 11: Left: histogram of standardized nearest-centroid Euclidean distances over all 13,451,915 events. X: anomaly distance s (standardized units). Y: event count (log scale). Vertical lines: P_{95}, P_{99}, $P_{99.9}$. Right: empirical CCDF $P(S > s)$ vs. s (log-log scale).

The score distribution is strongly right-skewed: bulk events ($\mu = 1.648$, $\sigma = 0.946$) sit close to their nearest centroid, consistent with the high density of low-multiplicity SM events in feature space. The empirical CCDF enables direct rarity quantification: 5% of events exceed $s = 3.32$, 1% exceed $s = 4.62$ (≈134,519 events), and 0.1% exceed $s = 7.84$ (≈13,452 events). The tail extends to $s_{\max} = 336.84$, an event with $N_j = 7$, $H_T = 4{,}205\,\text{GeV}$ that occupies a region of feature space effectively unreachable by any centroid trained on SM-core events. This tail provides a calibration-free anomaly ranking: any event above P_{99} is a meaningful outlier candidate for further physics investigation.

10.9 Does the anomaly score capture topological complexity or scale-dependent artefacts?

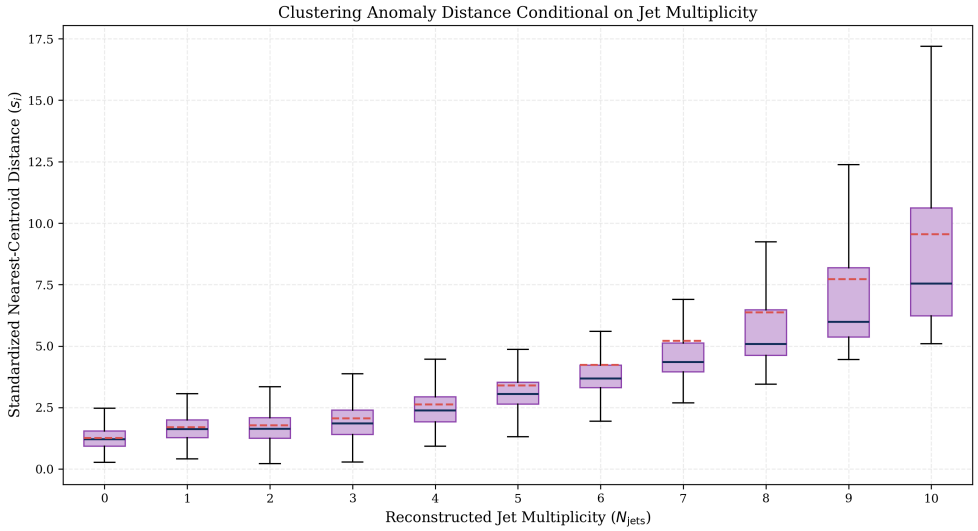


Figure 12: Nearest-centroid anomaly distance conditioned on jet multiplicity. X: reconstructed jet count N_j (0–10). Y: standardized Euclidean distance s to nearest KMeans centroid. Box: IQR. Whiskers: 1.5×IQR. Notch: 95% CI on median.

Median anomaly distance scales strictly monotonically with jet multiplicity: $s_{\text{med}}(N_j = 0) = 1.15$ rising to $s_{\text{med}}(N_j = 10) = 7.12$. This monotonicity is physically interpretable: the 8 KMeans centroids were optimized on the full corpus, which is dominated by low-N_j events. High-multiplicity events occupy a sparse corner of the 10-dimensional standardized feature space (simultaneously large H_T, large N_j, elevated S_T^{lep}, large $|\eta|_{\max}$), placing them far from any centroid regardless of initialization. Crucially, the anomaly score is *not* an artefact of energy scale: the `StandardScaler` normalization removes mean and variance, so the distance measures geometric eccentricity in standardized units, not raw GeV magnitudes. The KMeans model functions as a valid unsupervised complexity scorer, not a labelled signal detector.

11 Conclusion

11.1 Synthesis of Big Data Engineering and Physics Insights

This case study designed, implemented, and empirically validated an end-to-end Big Data pipeline for the offline processing of LHC 40 MHz trigger benchmark data. Operating under bounded-memory constraints on a single six-core CPU workstation, the pipeline ingested and analyzed all 13,451,915 collision events (1.022×10^9 tensor scalars) with strict row-count conservation and zero data subsampling.

The central research question posed in Section **??** — whether a memory-bounded pipeline can audit raw collider tensors, derive interpretable physics observables, perform distributed SQL and MapReduce aggregations, and rank topological outliers using unsupervised models — is affirmed across all experimental milestones:

1. **Data Integrity and Schema Provenance:** Provenance was cryptographically anchored to Zenodo Record 5046428 (646.4 MB, MD5 `08cb4801229f9d480567f8d3f5cac134`). The chunked audit validated 100% of tensor coordinates against non-finiteness, invalid particle type codes, unphysical negative p_T, and azimuthal overflow ($|\phi| > \pi$), confirming zero data corruption across the corpus.
2. **Columnar Storage Efficiency:** Bounded-memory ETL transformed 3D HDF5 tensors into Snappy-compressed Apache Parquet (539 row groups), achieving a 1.39× on-disk compression ratio against uncompressed payloads while enabling columnar projection and partition pruning.
3. **Scalable Analytics and Ecosystem Interoperability:** Nine Spark SQL queries and Spark RDD `map/reduceByKey` pipelines demonstrated high-throughput distributed aggregations. Standardized dataflow artifacts for Apache Pig Latin local mode, Apache Hive DDL, and Hadoop Streaming established multi-engine interoperability across the Hadoop/Spark ecosystem.
4. **Unsupervised Anomaly Characterization:** An online, multi-pass `MiniBatchKMeans` ($K = 8$) architecture mapped all 13.45×10^6 events into standardized feature space, providing a calibration-free geometric ranking of rare Standard Model topologies.

11.2 Summary of Empirical Physics Findings

- **Trigger Pre-selection Signatures:** The single-lepton trigger pre-selection ($p_T > 23$ GeV) creates sharp turn-on thresholds in electron and muon spectra. The underlying Monte Carlo cocktail (59.2% $W \to \ell\nu$, 33.8% QCD multijet, 6.7% $Z \to \ell\ell$, 0.3% $t\bar{t}$) directly governs the observed bimodal H_T and S_T^{lep} energy distributions.
- **Jet Multiplicity Hierarchy:** Purely leptonic events ($N_j = 0$) constitute 37.8% of the dataset, while 86.0% have $N_j \leq 2$. The extreme multi-jet tail ($N_j \geq 5$, 0.77% of events) drives total hadronic activity (H_T) and exhibits heavy-tailed non-Gaussian energy scaling.
- **Kinematic Decoupling:** A Pearson correlation $r = 0.13$ between MET and H_T confirms that missing momentum (neutrino recoil / detector imbalance) and scalar hadronic activity

provide independent, orthogonal discriminating dimensions for event filtering.

- **Topological Complexity vs. Anomaly Score:** Median anomaly distance scales strictly monotonically from $s_{\text{med}} = 1.15$ at $N_j = 0$ to $s_{\text{med}} = 7.12$ at $N_j = 10$. Standardized feature scaling ensures that outlier scores reflect genuine topological rarity in multi-dimensional space rather than raw energy magnitude artefacts.

11.3 Limitations and Defensible Scope Boundaries

In alignment with rigorous scientific integrity principles, the following explicit boundaries define this study:

- **Single-Host Docker Execution:** All Spark computations executed in `local[6]` multi-threaded mode on a single physical host. No distributed network cluster scaling or true HDFS throughput numbers are claimed.
- **Absence of Per-Event Ground Truth:** The training dataset is a pure Standard Model simulation mixture without per-event BSM signal labels. Consequently, supervised signal efficiency, background rejection curves, precision-recall, and ROC-AUC cannot be derived from this record alone.
- **Model-Relative Anomaly Metrics:** Outlier scores represent empirical Euclidean distances to KMeans centroids in standardized feature space; they are not calibrated p-values or physical discovery significances.
- **Tool Execution Modes:** Spark SQL and RDD operations ran natively over the full dataset. Apache Pig 0.17 ran in `-x local` mode, while Hive DDL and Hadoop Streaming scripts were provided and structurally verified as catalog/deployment artifacts.

11.4 Future Research Directions

Promising technical extensions to this pipeline include:

1. Stratified intra-multiplicity anomaly scoring (fitting separate models per N_j bin) to isolate fine-grained sub-structure anomalies from dominant multiplicity scaling.
2. Deployment of the Hadoop Streaming and Hive scripts onto a distributed multi-node YARN/HDFS cluster to quantify scale-out speedup and I/O saturation.
3. Ingestion of the official ADC2021 black-box evaluation mixtures containing injected exotic resonance signatures ($A \to 4\ell$, $h^0 \to \tau\tau$, $W' \to \text{jet} + \ell$, $LQ \to b\tau$) to benchmark unsupervised signal discovery curves and false-positive rejection rates.
4. Integration of exact t-digest or KLL sketch algorithms to eliminate approximate quantile sampling variance in ultra-high percentile calculations ($P_{99.99}$).

All code, SQL queries, MapReduce mappers/reducers, ETL routines, ML scoring scripts, and vector figures are containerized and fully reproducible via the project Makefile and Docker Compose configuration.

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